

//SIT 55 : ASSIGNMENT 9 : CASE STUDY

/*Using concepts of Object-Oriented programming develop solution for any one application

1) Banking system having following operations :

1. Create an account
2. Deposit money
3. Withdraw money
4. Honor daily withdrawal limit
5. Check the balance
6. Display Account information.

2) Inventory management system having following operations :

1. List of all products
2. Display individual product information
3. Purchase
4. Shipping
5. Balance stock
6. Loss and Profit calculation.*/*

```
import java.util.Scanner;
class Customer
{
    private String customerName;
    private int customerAge;
    public void setCustomerName(String customerName)
    {
        this.customerName=customerName;
    }
    public String getCustomerName()
    {
        return customerName;
    }
    public void setCustomerAge(int customerAge)
    {
        this.customerAge=customerAge;
    }
    public int getCustomerAge()
    {
        return customerAge;
    }
}
abstract class Account
{
    protected double balance;
    protected int accountId;
    protected String accountType;
    protected Customer custobj;
    void setBalance(double balance)
    {
        this.balance=balance;
    }
    double getBalance()
    {
        return balance;
    }
    void setAccountId(int accountId)
    {
        this.accountId=accountId;
    }
    int getAccountId()
    {
        return accountId;
    }
    void setAccountType(String accountType)
```

```

{
    this.accountType=accountType;
}
String getAccountType()
{
    return accountType;
}
void setCustomerObject(Customer custobj)
{
    this.custobj=custobj;
}
Customer getCustomerObject()
{
    return custobj;
}
public abstract boolean withdraw(double amount);
}

```

//SAVING ACCOUNT CLASS

```

class SavingsAccount extends Account
{
    private double minimumBalance;
    public void setMinimumBalance(double minimumBalance)
    {
        this.minimumBalance=minimumBalance;
    }
    public double getMinimumBalance()
    {
        return minimumBalance;
    }
    public boolean withdraw(double amount)
    {
        if((balance-amount)>minimumBalance)
        {
            balance-=amount;
            return true;
        }
        else
            return false;
    }
}

```

//BANK CLASS

```

class Bank
{
    public static Scanner sc=new Scanner(System.in);
    public SavingsAccount a=new SavingsAccount();
    public Customer c=new Customer();
    public SavingsAccount createAccount()
    {
        System.out.print("ENTER YOUR NAME:");
        String customername=sc.nextLine();
        c.setCustomerName(customername);
        System.out.print("ENTER YOUR AGE: ");
        int customerage=sc.nextInt();
    }
}

```

```

if(customerage<18)
{
    do
    {
        System.out.print("MINIMUM AGE SHOULD BE 18 TO CREATE AN ACCOUNT.\nPLEASE ENTER VAL
ID AGE: ");
        customerage=sc.nextInt();
    }while(customerage<18);
    }
    c.setCustomerAge(customerage);
    a.setCustomerObject(c);
    System.out.print("ENTER YOUR ACCOUNT ID: ");
    int accountid=sc.nextInt();
    a.setAccountid(accountid);
    System.out.print("ENTER YOUR ACCOUNT TYPE: ");
    String accounttype=sc.next();
    a.setAccountType(accounttype);
    System.out.print("ENTER BALANCE: ");
    double balance=sc.nextDouble();
    a.setBalance(balance);
    System.out.print("ENTER MINIMUM BALANCE: ");
    double minbalance=sc.nextDouble();
    a.setMinimumBalance(minbalance);
    return a;
}
void getWithdrawAmount()
{
    System.out.print("ENTER THE AMOUNT YOU WANT TO WITHDRAW: ");
    double amount=sc.nextDouble();
    if(amount>20000)
    {
        System.out.println("WITHDRAWL FAILED.MAXIMUM LIMIT OF WITHDRAWL IN ONE TRANSACTION
IS RS.20000");
    }
    else
    {
        if(a.withdraw(amount)==true)
        {
            System.out.println("WITHDRAWL SUCCESSFUL. BALANCE IS: "+a.getBalance());
        }
        else
        {
            System.out.println("SORRY!! NOT ENOUGH BALANCE");
        }
    }
}
public void depositAmount(double amount)
{
    double bal=a.getBalance()+amount;
    a.setBalance(bal);
    System.out.println("AMOUNT DEPOSITED SUCCESSFULLY. BALANCE IS: "+a.getBalance());
}
public void checkBalance()
{
    System.out.println("BALANCE IS: "+a.getBalance());
}

```

```
}
public void displayAccountInformation()
{
    System.out.println("WELCOME "+c.getCustomerName()+"! FOLLOWING ARE YOUR ACCOUNT DETAILS:");
    System.out.println("AGE :"+c.getCustomerAge());
    System.out.println("ACCOUNT ID: "+a.getAccountId());
    System.out.println("ACCOUNT TYPE: "+a.getAccountType());
    System.out.println("BALANCE: "+a.getBalance());
    System.out.println("MINIMUM BALANCE: "+a.getMinimumBalance());
}
}
```

```

public class Main
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        SavingsAccount a;
        Bank bm=new Bank();
        do
        {
            System.out.println("\n1.CREATE ACCOUNT \n2.DISPLAY ACCOUNT \n3.CHECK BALANCE"+
            "\n4.DEPOSIT AMOUNT \n5.WITHDRAW AMOUNT \n6.EXIT \n");
            System.out.print("ENTER YOUR CHOICE: ");
            int choice=sc.nextInt();
            System.out.println("");
            switch(choice)
            {
                case 1:
                    a=bm.createAccount();
                    System.out.println("=====");
                    break;
                case 2:
                    bm.displayAccountInformation();
                    System.out.println("=====");
                    break;
                case 3:
                    bm.checkBalance();
                    System.out.println("=====");
                    break;
                case 4:
                    System.out.print("ENTER AMOUNT YOU WANT TO DEPOSIT: ");
                    double amount=sc.nextDouble();
                    bm.depositAmount(amount);
                    System.out.println("=====");
                    break;
                case 5:
                    bm.getWithdrawAmount();
                    System.out.println("=====");
                    break;
                case 6:
                    System.out.println("=====");
                    return ;
                default:
            }
        }
    }
}

```

```
        System.out.println("INVALID INPUT !!");
        System.out.println("=====");
        break;
    }
}while(true);

}
}
```

/*OUTPUT:

1.CREATE ACCOUNT
2.DISPLAY ACCOUNT
3.CHECK BALANCE
4.DEPOSIT AMOUNT
5.WITHDRAW AMOUNT
6.EXIT

ENTER YOUR CHOICE: 1

ENTER YOUR NAME:ABC
ENTER YOUR AGE: 20
ENTER YOUR ACCOUNT ID: 551155
ENTER YOUR ACCOUNT TYPE: SAVING
ENTER BALANCE: 5500
ENTER MINIMUM BALANCE: 1000

=====

1.CREATE ACCOUNT
2.DISPLAY ACCOUNT
3.CHECK BALANCE
4.DEPOSIT AMOUNT
5.WITHDRAW AMOUNT
6.EXIT

ENTER YOUR CHOICE: 2

WELCOME ABC! FOLLOWING ARE YOUR ACCOUNT DETAILS:

AGE :20
ACCOUNT ID: 551155
ACCOUNT TYPE: SAVING
BALANCE: 5500.0
MINIMUM BALANCE: 1000.0

=====

1.CREATE ACCOUNT
2.DISPLAY ACCOUNT
3.CHECK BALANCE
4.DEPOSIT AMOUNT
5.WITHDRAW AMOUNT
6.EXIT

ENTER YOUR CHOICE: 3

BALANCE IS: 5500.0

=====

- 1.CREATE ACCOUNT
- 2.DISPLAY ACCOUNT
- 3.CHECK BALANCE
- 4.DEPOSIT AMOUNT
- 5.WITHDRAW AMOUNT
- 6.EXIT

ENTER YOUR CHOICE: 4

ENTER AMOUNT YOU WANT TO DEPOSIT: 3300
AMOUNT DEPOSITED SUCCESSFULLY. BALANCE IS: 8800.0

=====

- 1.CREATE ACCOUNT
- 2.DISPLAY ACCOUNT
- 3.CHECK BALANCE
- 4.DEPOSIT AMOUNT
- 5.WITHDRAW AMOUNT
- 6.EXIT

ENTER YOUR CHOICE: 5

ENTER THE AMOUNT YOU WANT TO WITHDRAW: 2500
WITHDRAWL SUCCESSFUL. BALANCE IS: 6300.0

=====

- 1.CREATE ACCOUNT
- 2.DISPLAY ACCOUNT
- 3.CHECK BALANCE
- 4.DEPOSIT AMOUNT
- 5.WITHDRAW AMOUNT
- 6.EXIT

ENTER YOUR CHOICE: 6

=====*/